

JACOB IWEN

Senior Full Stack Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 8+ years building and operating enterprise software, from application design through deployment and production support. Core engineer on a multi-tenant form-digitization SaaS product from its initial buildout, with long-term ownership of Rails portals, ETL pipelines, document workflows, and AWS infrastructure. Experience spans TypeScript, Node.js, React, Ruby, and relational databases.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Ruby, Python, SQL, Bash
Backend	Node.js, Express, Firebase Cloud Functions, Ruby on Rails, REST API design, Resque/Redis job systems, Puppeteer PDF generation (headless Chromium)
Frontend	React (TypeScript, hooks), Angular, AngularJS, Vite and custom webpack build pipelines, styled-components
Cloud and Ops	AWS (EC2, RDS, VPC, S3, CloudWatch), GCP/Firebase, Linux administration (nginx, systemd, cron)
Data and Integrations	MySQL, SQLite, ETL pipelines (SFTP, Excel/CSV, fixed-width), X12 EDI (837P/835/999), Twilio, webhooks, Quickbase
Auth and Security	OAuth 2.0, JWT, magic-link authentication, 2FA token infrastructure, TLS/CSP hardening
Testing and Delivery	Jest, RSpec, Selenium, GitHub Actions CI/CD, staged dev/test/prod releases
AI Integration	MCP servers, OpenAI Agents SDK, Codex app-server integration, LLM product features, AI code-review workflows (Claude Code, Codex), 2025–2026

PROFESSIONAL EXPERIENCE

Advantage Integrated Solutions | Senior Full Stack Engineer

March 2018 to July 2026

Solution Developer to Senior Full Stack Engineer (promoted May 2026)

Custom software consultancy serving 15+ enterprise clients, including BASF, Intuit, VMware, and BrightSpring/ResCare. Worked directly with clients to understand business and data requirements, recommend technical approaches, and own production systems across frontend, backend, data, and infrastructure.

Form Engine | Multi-tenant form-digitization SaaS (2018–2026)

- One of three core engineers on the product for eight years. Built the form-builder core during the initial 10-week buildout and owned the Puppeteer platform for pixel-accurate PDF generation. The product combined an AngularJS application, TypeScript Firebase Cloud Functions, and e-signatures. Data providers included Quickbase, ServiceNow, and Salesforce.
- Built LLM-based form-template generation in 2026 using MCP tools and the OpenAI Agents SDK, with transaction logging for auditability. Introduced Claude Code PR-assistant and AI code-review workflows in GitHub Actions.
- Replaced full-template API requests with compact, endpoint-specific payloads, eliminating multi-second uploads in Fast 3G tests. Optimized template fetching, roughly halving page-load and template-fetch times in development benchmarks. Reduced the JavaScript bundle size by 25%.
- Independently built an X12 healthcare EDI module for 837P claims and 835 remittances, with SFTP delivery to the federal WCMBP portal, enabling a home-healthcare client to bill a federal workers' compensation program.
- Led Node.js 14 and 22 runtime migrations and implemented a migration from the legacy XML API to JSON REST endpoints.

Client Platform Ownership | Advantage Integrated Solutions (continued)

- Long-term primary contributor to BrightSpring/ResCare's production ETL and background-job platform: 50+ jobs across roughly 30 active cron schedules integrating state and county workforce agencies through SFTP and Excel/CSV pipelines. Independently migrated the server to Amazon Linux 2023 and Ruby 3.3, and built Firebase-backed authentication-ticket renewal when Quickbase enforced 2FA.
- Owned BASF/Thurston Genetics' customer-facing seed-royalty portal for more than seven years. Rebuilt royalty and forecasting forms for reporting three times a year, introduced webpack, and delivered Ruby/Rails upgrades and an AWS RDS migration.
- Created react-exact-form as its sole author: a React framework and scaffolder for 30+ pixel-accurate, print-ready forms across 5+ clients. Built a separate single-file React deployment toolchain used across dozens of regional client pages.
- Owned a home-healthcare client's nightly Quickbase-to-MySQL archive from 2022 to 2026. Audited MySQL 8.4 compatibility and designed a column-rename queue that avoided MySQL's column limit and table rebuilds. Contributed to clinical shift-conflict detection that handled time zones and daylight saving time.
- Sole maintainer of Intuit HR's employee-separation document system from 2023 to 2026, supporting eight country-specific templates. For a separate insurance client, delivered OSHA 300/301 workflows, Content Security Policy and HTTPS enforcement, and a self-healing Resque queue monitor.

Infrastructure, Security, and Technical Leadership

- Operated a self-managed AWS fleet with EC2 AMI/snapshot backups, failure alerts, RDS migrations, uptime monitoring, and SSL rollout tooling with automatic rollback across six servers. Handled production incidents, from emergency patches for breaking vendor API changes to rebuilding a client's deleted pages.
- Kept production email working as Google retired password-based SMTP access by building a Python OAuth 2.0 toolkit and migrating email services to service accounts with domain-wide delegation. Also redacted sensitive data from failure notifications and fixed a credential leak in an authentication redirect.
- Reviewed SSO, WebAuthn, and JWT changes by testing branches and contributing fixes to teammates' pull requests. Wrote onboarding and operations documentation, set code standards for a shared Cloud Functions codebase, and built drop-in widgets so non-engineer consultants could add PDF generation to client apps.

Creed Interactive | Contract Developer

January 2018 to March 2018

- Automated a Salesforce catalog migration with Python and JavaScript parsers that transformed legacy HTML/XML into structured XML/CSV while preserving SEO.

BounceTribe | Full Stack Developer (freelance)

August 2017 to March 2018

- Built a musician collaboration platform with a React front end and GraphQL/Express backend, including JWT authentication, Mailgun email, and AWS S3 uploads.

SELECTED PROJECTS

- **Fieldwork | Job-search application (2026):** Built a React 19/TypeScript and Node.js application that runs Codex agent sessions inside a sandboxed draft with captured source snapshots and defined file tools, rejects stale edits with source-hash checks, and validates generated PDFs before download.
- **Vehicle diagnostics and telemetry platform (2026):** Built a Python CAN/UDS platform on Raspberry Pi with a browser dashboard, SQLite trip history, and synchronized three-bus recording. Restored adaptive cruise control by physically aligning the radar and executing an undocumented calibration routine.
- **Vehicle network and automation stack (2019–present):** Built and operate an OpenWrt/Raspberry Pi stack with a React dashboard, vehicle ignition-aware HDD storage protection, automated backups, and WAN failover across directional Wi-Fi, Starlink, and LTE.
- **Open source:** Fixed Windows-1252/UTF-8 encoding in node-quickbase; the patch was accepted upstream.

EDUCATION

- **Full Stack Software Engineering Certification**, PRIME Digital Academy, 2017
- **B.S. Mechanical Engineering**, University of Minnesota, Twin Cities, 2012